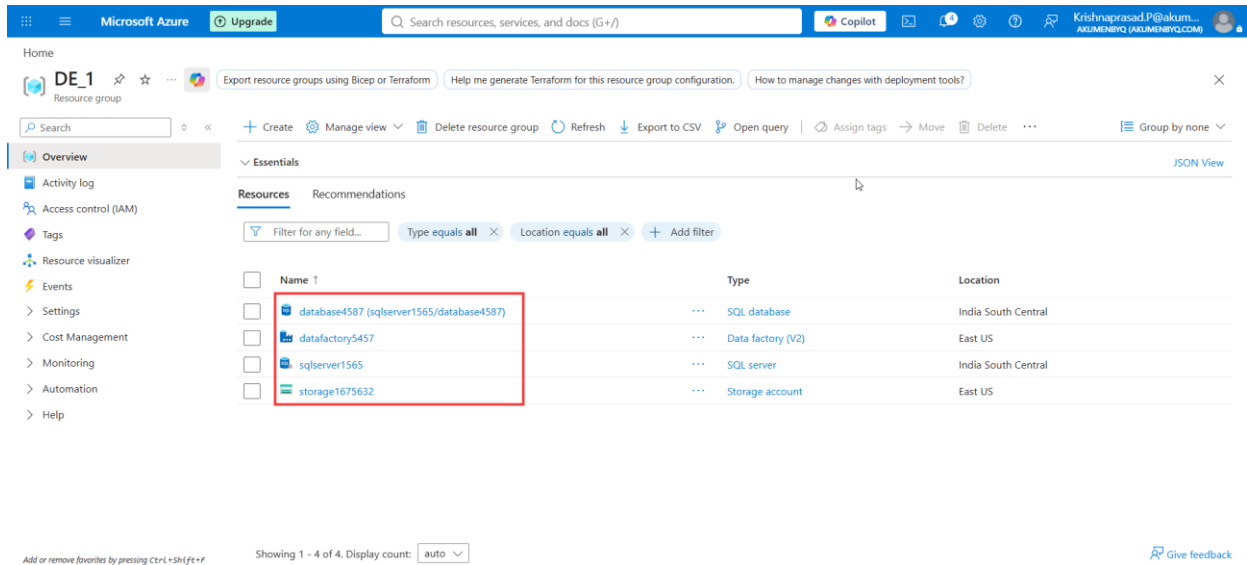


Retail Data Integration Platform — ADF Build

Step 1: Verify the Resource Group

- Opened resource group DE_1, containing four resources: database4587 (SQL database), datafactory5457 (Data Factory V2), sqlserver1565 (SQL server) and storage1675632 (Storage account).

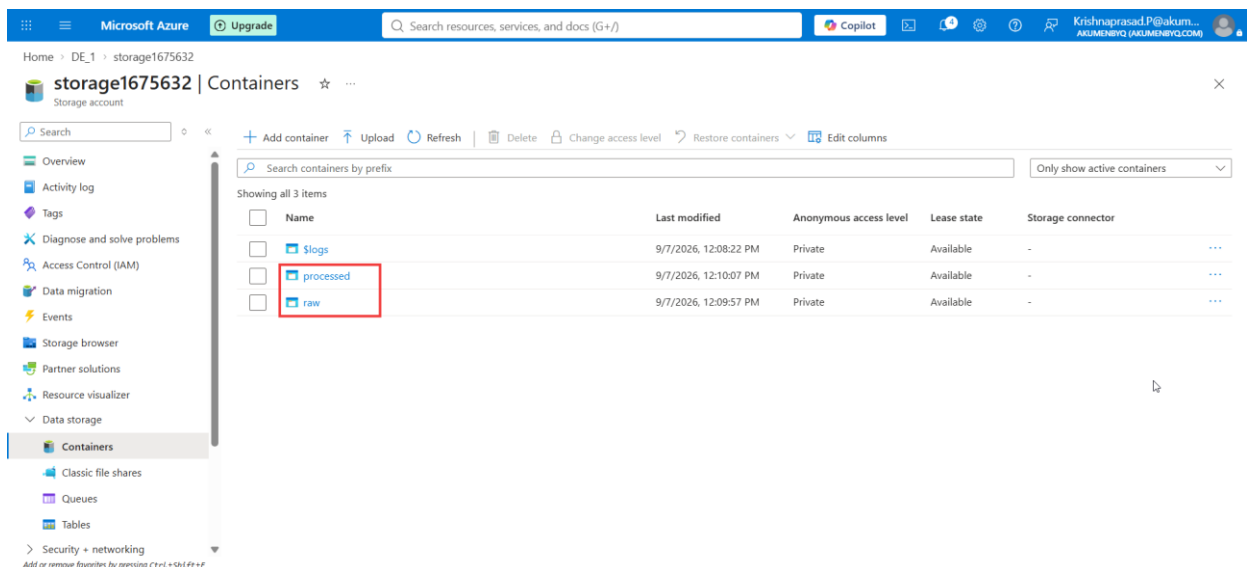


The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and a Copilot icon. The user's profile is visible in the top right corner. The main content area displays the 'DE_1' resource group. The 'Resources' tab is selected, showing a table of resources. A red box highlights the following resources:

Name	Type	Location
database4587 (sqlserver1565/database4587)	SQL database	India South Central
datafactory5457	Data factory (V2)	East US
sqlserver1565	SQL server	India South Central
storage1675632	Storage account	East US

Step 2: Create the Bronze and Silver Zone Containers

- In storage1675632, created the raw container (Bronze zone) and processed container (Silver zone), alongside the existing \$logs container.

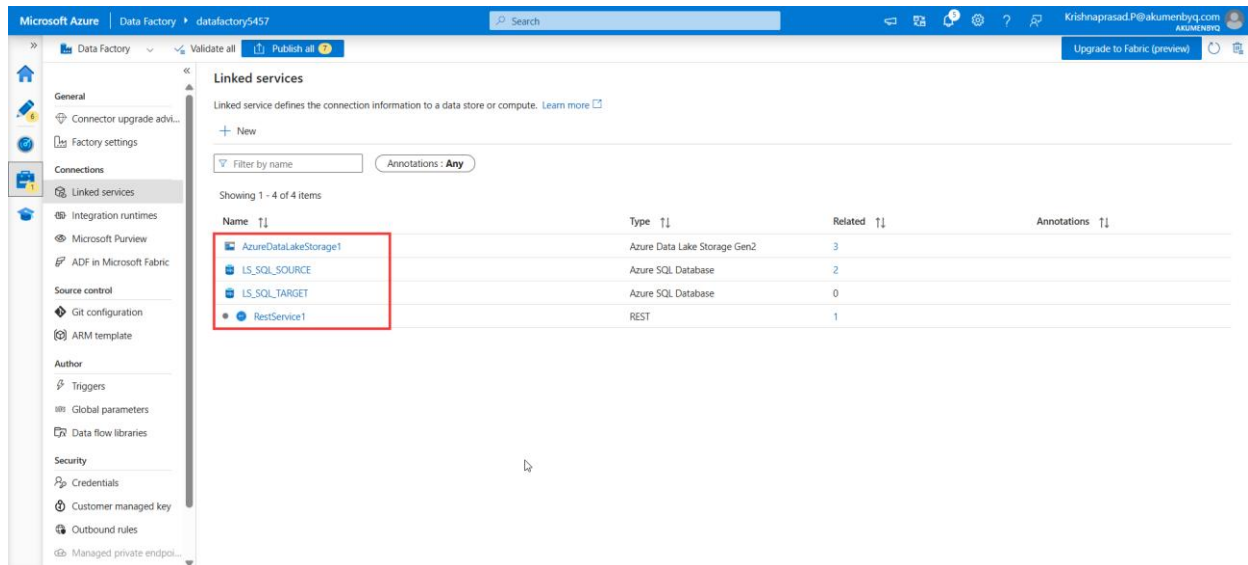


The screenshot shows the Microsoft Azure portal interface for the 'storage1675632' storage account. The 'Containers' tab is selected, displaying a table of containers. A red box highlights the 'processed' and 'raw' containers.

Name	Last modified	Anonymous access level	Lease state	Storage connector
\$logs	9/7/2026, 12:08:22 PM	Private	Available	-
processed	9/7/2026, 12:10:07 PM	Private	Available	-
raw	9/7/2026, 12:09:57 PM	Private	Available	-

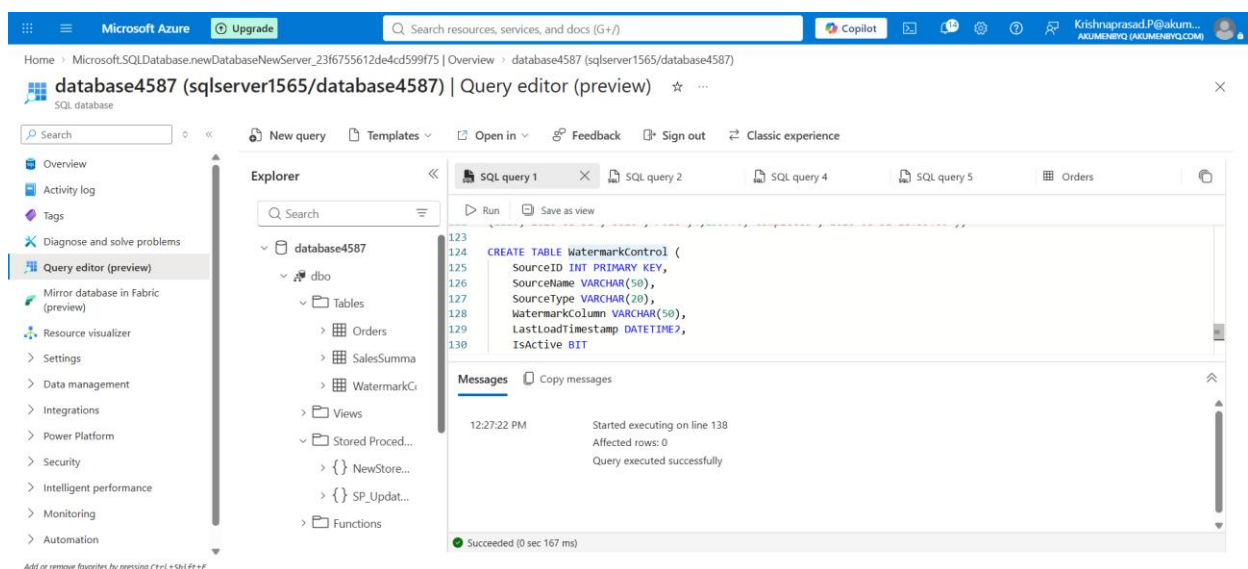
Step 3: Create the Linked Services

- In Data Factory, created four linked services: AzureDataLakeStorage1 (Bronze/Silver zone storage), LS_SQL_SOURCE (Orders source database), LS_SQL_TARGET (curated database) and RestService1 (vendor Product Catalog REST API).



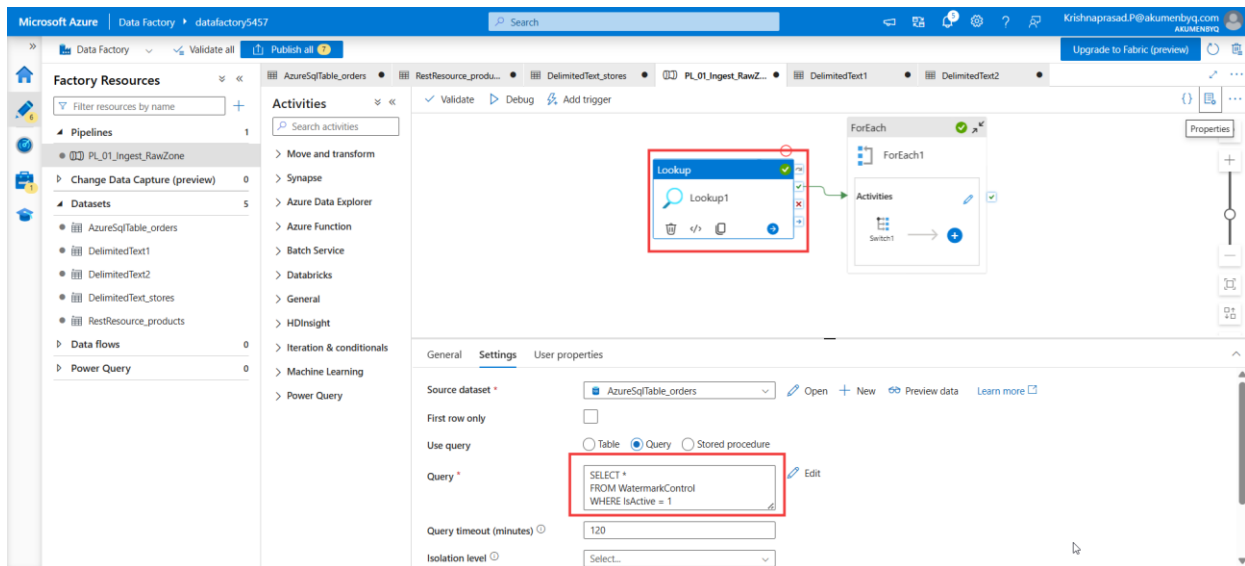
Step 4: Create Tables in SQL

- In the source SQL database, ran a script to create the Orders, WatermarkControl, SalesSummary table — the metadata-driven control table that lets PL_01 loop over each source without hardcoding pipeline logic.

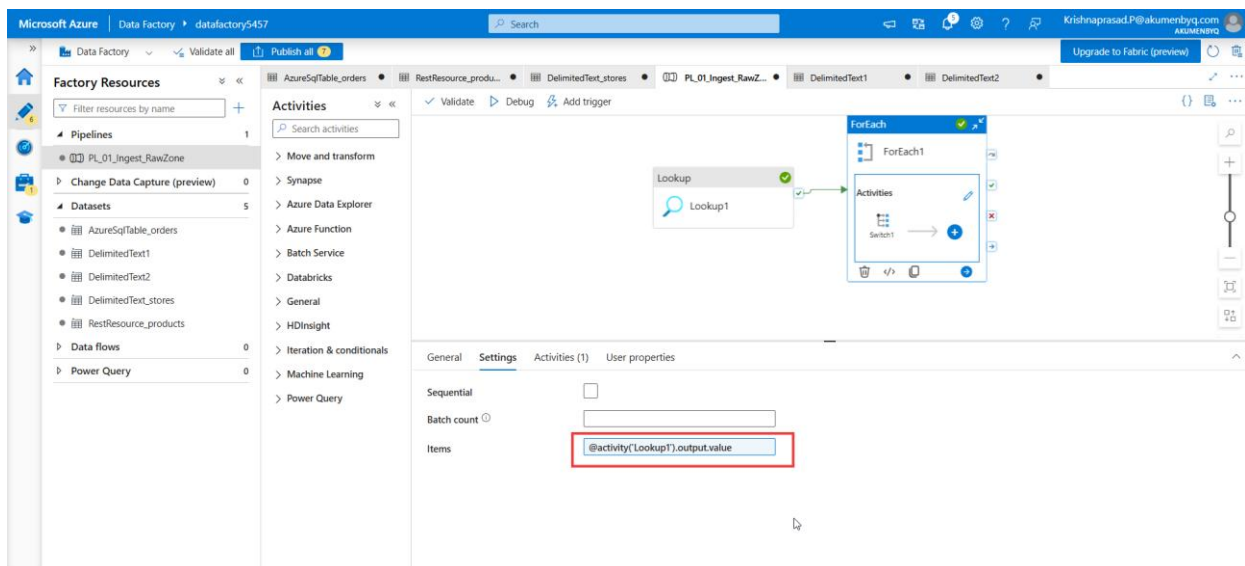


Step 5: Build PL_01_Ingest_RawZone — Look Up Active Sources

- Created pipeline PL_01_Ingest_RawZone and added a Lookup activity (Lookup1) querying `SELECT * FROM WatermarkControl WHERE IsActive = 1`, returning the full row set so every active source can be iterated over.

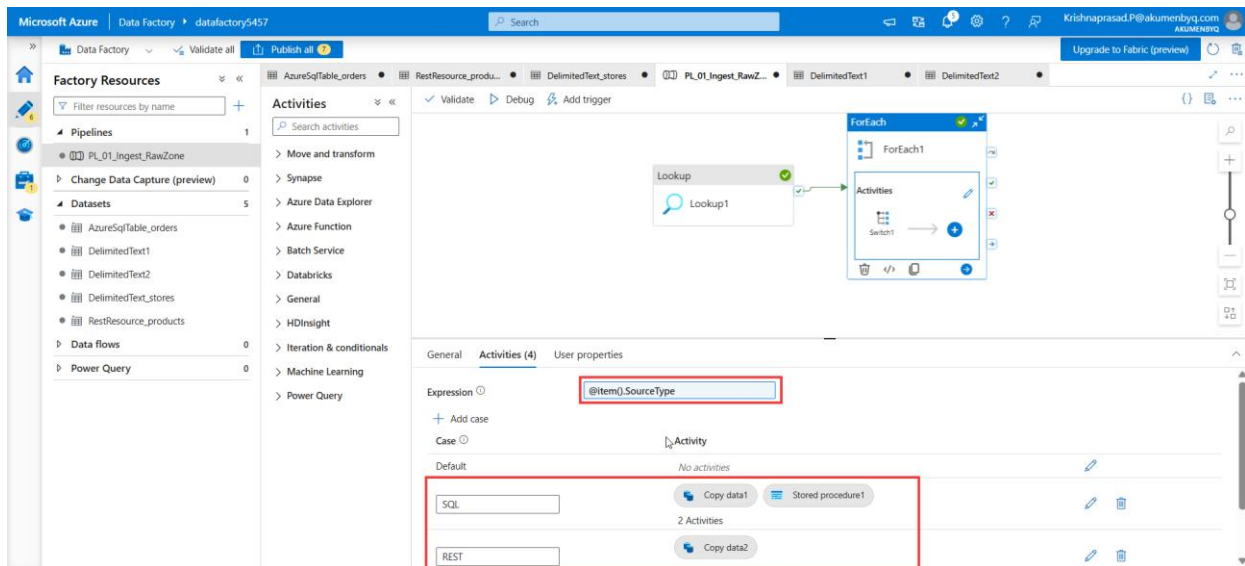


- Added a ForEach activity after the Lookup, with Items set to `@activity('Lookup1').output.value` so the loop runs once per active source row.

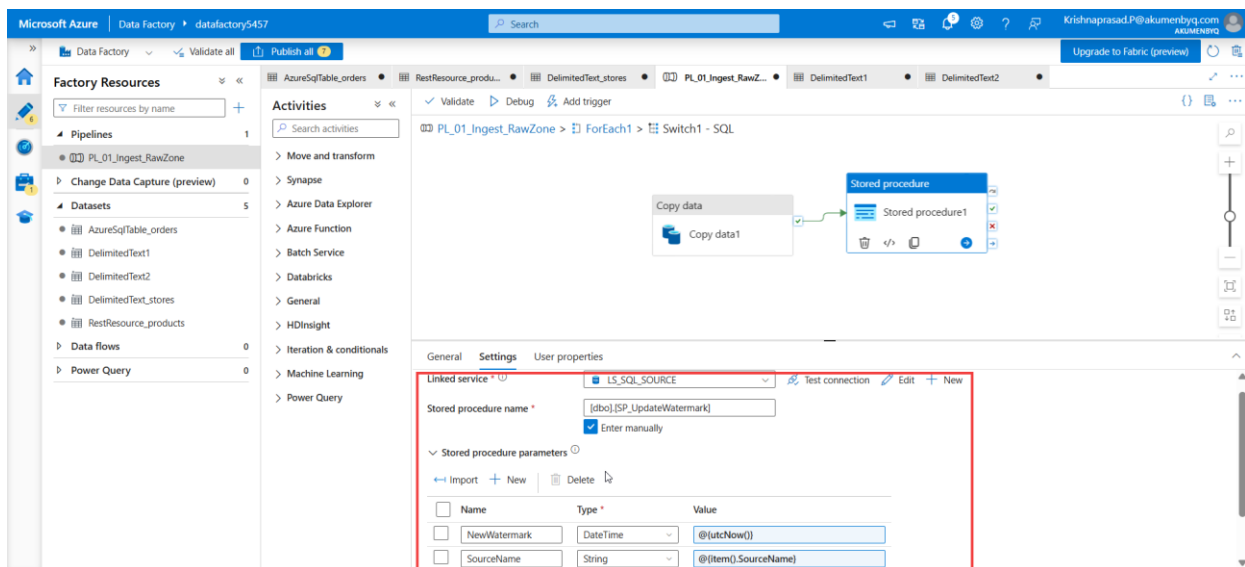


Step 6: Branch Ingestion Logic by Source Type

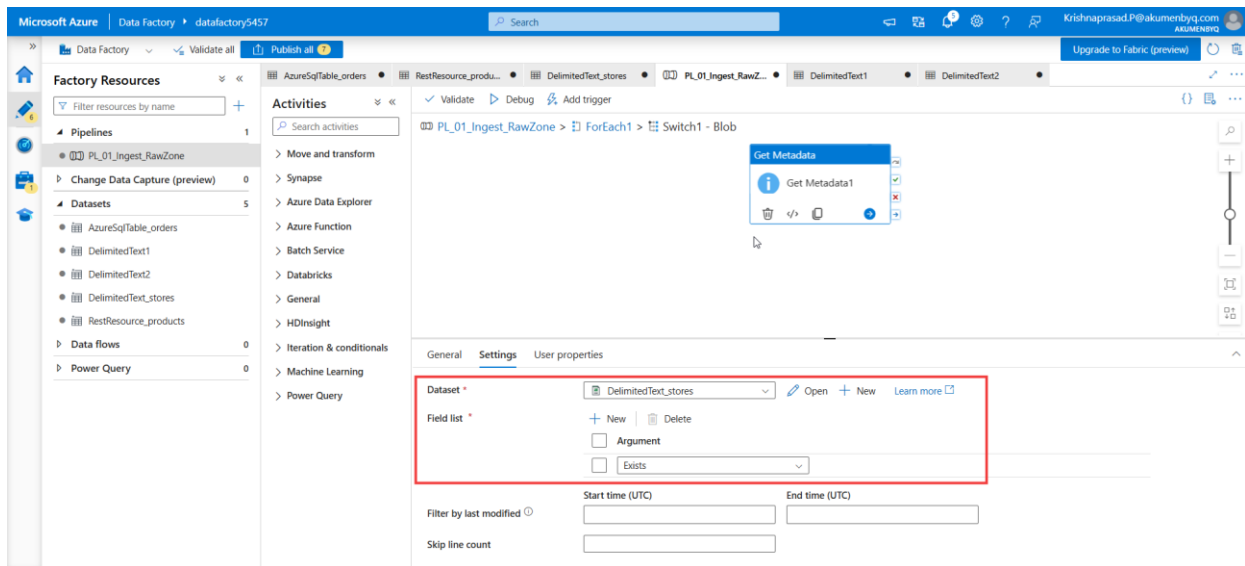
- Inside the ForEach loop, added a Switch activity (Switch1) on the expression `@item().SourceType`, with a SQL case (Copy data1 + Stored procedure1), a REST case (Copy data2) and no Default activity — routing each source to the right ingestion pattern.



- In the SQL case, after Copy data1 lands the Orders extract, Stored procedure1 calls [dbo].[SP_UpdateWatermark] via linked service LS_SQL_SOURCE, passing NewWatermark = @utcNow() and SourceName = @item().SourceName to advance the watermark for the next incremental run.

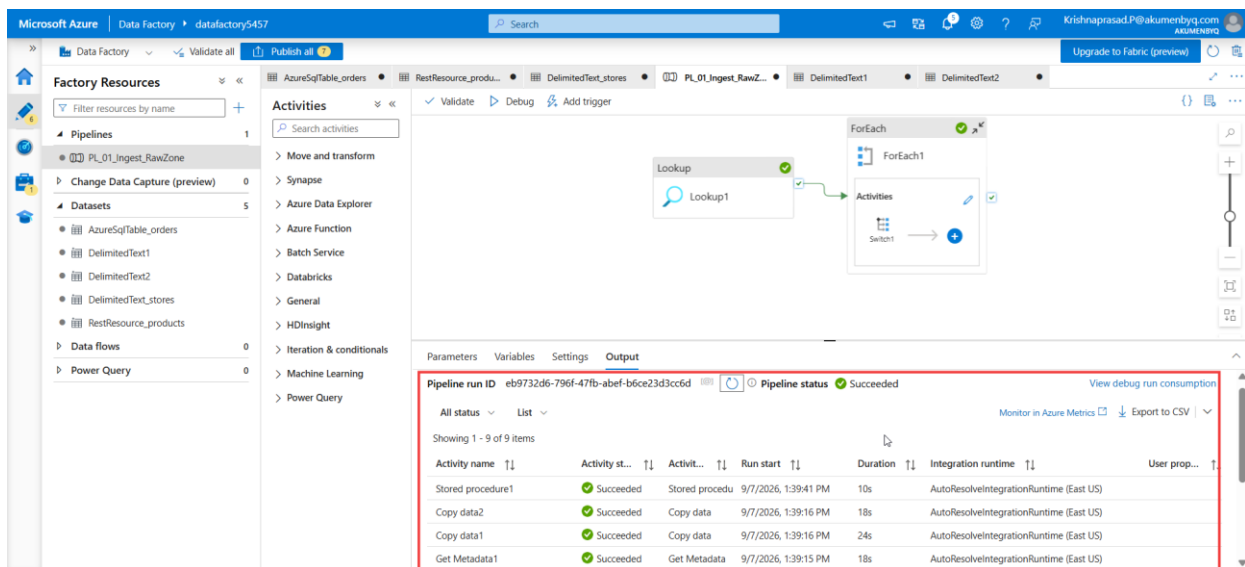


- In the Blob case, a Get Metadata activity checks the DelimitedText_stores dataset for the Exists field, confirming the Store Master CSV has arrived before it's processed.



Step 7: Run and Verify the Ingestion Pipeline

- Ran Debug on PL_01_Ingest_RawZone — all activities across the three ForEach iterations succeeded: Stored procedure1, Copy data2, Copy data1 and Get Metadata1.



- Verified the raw container: Products.csv, Stores.csv and dbo.Orders.csv were all landed in the Bronze zone, one file per source.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > storage1675648 | Containers

raw ...

Container

Search

+ Add Directory | Upload | Refresh | Delete | Copy | Paste | Rename | Acquire lease | Break lease | Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 3 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Products.csv	9/8/2026, 11:25:27 AM	Hot (Inferred)	Block blob	1.34 KiB	Available
Stores.csv	9/8/2026, 10:25:27 AM	Hot (Inferred)	Block blob	676 B	Available
dbo.Orders.csv	9/8/2026, 11:25:36 AM	Hot (Inferred)	Block blob	10.6 KiB	Available

Add or remove favorites by pressing Ctrl+Shift+F

Step 8: Build PL_02_Transform_ProcessedZone — Filter Orders

- Created pipeline PL_02_Transform_ProcessedZone with a Mapping Data Flow (dataflow1) reading orders, products and stores from the Bronze zone, then added a Filter transformation (filter1) removing rows with a null OrderID, ProductID, StoreID or OrderDate.

Microsoft Azure | Data Factory | datafactoryde265487

Search

Factory Resources

Filter resources by name

Pipelines

- PL_01_Ingest_RawZone

Datasets

- AzureSqlTable1_Orders
- DelimitedText1
- DelimitedText2
- DelimitedText3
- DelimitedText_Stores
- Orders
- Products
- RestResource_Products
- Stores

Data flows

- dataflow1

Power Query

0

Validate all | Publish all

Settings | Upgrade to Fabric (preview)

DelimitedText1 | PL_01_Ingest_RawZone | dataflow1 | DelimitedText2 | Orders | Products | Stores

Validate | Data flow debug | Debug Settings

Import data from Orders

filter1

Columns: 8 total

Import data from Products

products

Import data from Stores

stores

derivedColumn1

Creating/updating the columns 'OrderID, OrderDate, StoreID, and product'

join1

Inner join on 'derivedColumn1' and 'product'

join2

Inner join on 'join1' and 'stores'

aggregate1

Aggregating data by ProductID, StoreID, OrderDate, producing

Columns: 6 total

Filter settings | Optimize | Inspect | Data preview

Output stream name *

filter1

Learn more

Description

Filtering rows using expressions on columns: 'OrderID, ProductID, StoreID, OrderDate'

Reset

Incoming stream *

orders

Filter on *

IsNotNull(OrderID) && IsNotNull(ProductID) && IsNotNull(StoreID) && IsNotNull(OrderDate)

- Added a Derived Column transformation (derivedColumn1) standardizing OrderDate to yyyy-MM-dd and stamping a LoadDate of currentDate() on every row.

The screenshot shows the 'Derived column's settings' pane in Microsoft Azure Data Factory. The 'Columns' section is highlighted with a red box, displaying a table with columns and expressions:

Column	Expression
OrderDate	toDate(OrderDate, 'yyyy-MM-dd')
LoadDate	currentDate()

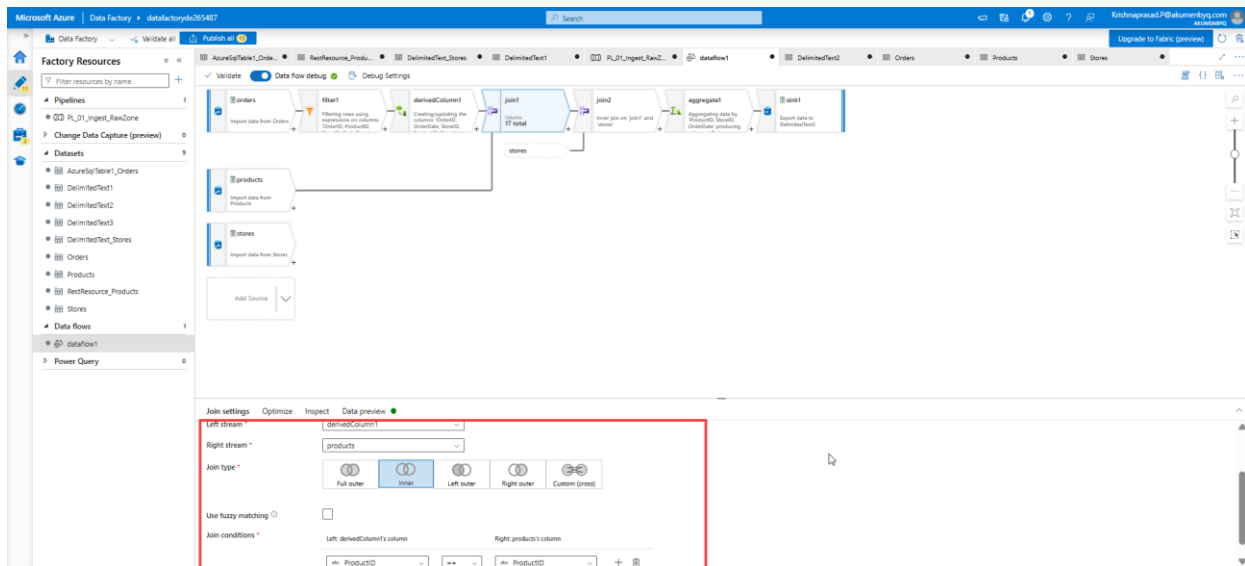
- Previewed the result — 100 rows with clean OrderDate, Quantity, UnitPrice, OrderStatus and LoadDate columns.

The screenshot shows the 'Data preview' pane in Microsoft Azure Data Factory. The preview table is highlighted with a red box, displaying 100 rows of data:

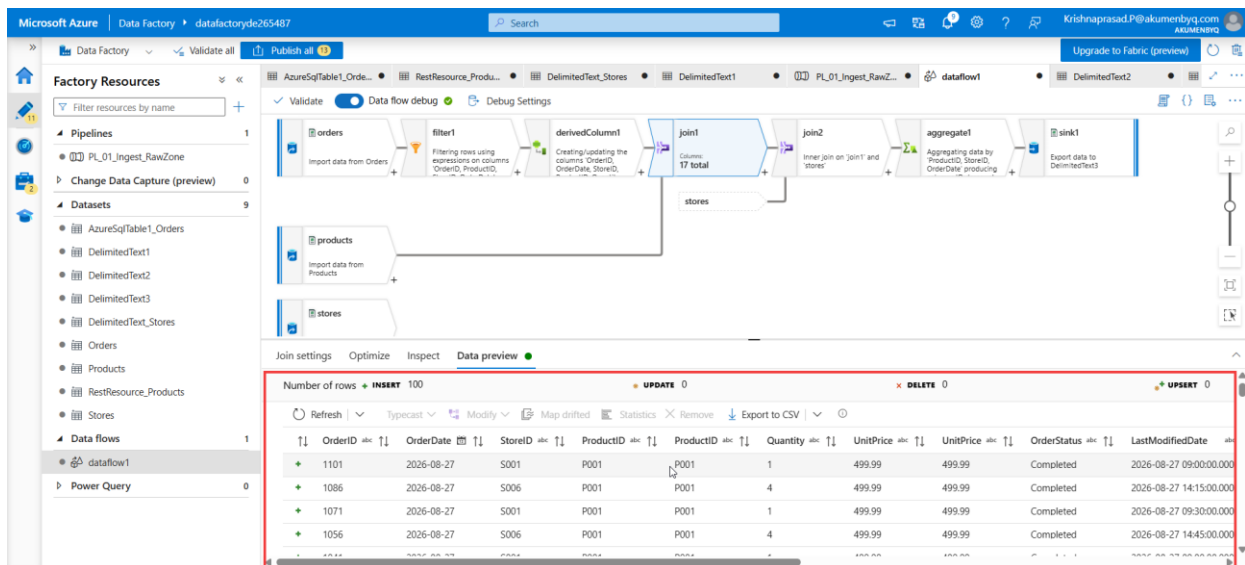
OrderID	OrderDate	StoreID	ProductID	Quantity	UnitPrice	OrderStatus	LastModifiedDate	LoadDate
1001	2026-08-27	S001	P001	1	499.99	Completed	2026-08-27 09:00:00.0000000	2026-09-08
1002	2026-08-28	S002	P002	2	2499.00	Completed	2026-08-28 10:07:00.0000000	2026-09-08
1003	2026-08-29	S003	P003	3	699.00	Completed	2026-08-29 11:14:00.0000000	2026-09-08
1004	2026-08-30	S004	P004	4	1299.00	Completed	2026-08-30 12:21:00.0000000	2026-09-08
1005	2026-08-27	S005	P005	5	3499.00	Cancelled	2026-08-27 13:28:00.0000000	2026-09-08

Step 9: Join Orders with Products

- Added an Inner Join (join1) between derivedColumn1 and products on ProductID.



- Previewed the joined output — 100 rows now enriched with product-side columns.



Step 10: Join with Stores

- Added a second Inner Join (join2) between join1 and stores on StoreID, bringing store details onto every order row.

Join settings

Left stream * join1

Right stream * stores

Join type * Inner

Use fuzzy matching ☐

Join conditions *

Left: join1's column Right: stores's column

StoreID == StoreID

Data preview

Columns: 24 total

- Previewed the result — 24 total columns across the combined orders, products and stores data.

Data preview

Number of rows: INSERT 100 UPDATE 0 DELETE 0

OrderID	OrderDate	StoreID	ProductID	Quantity	UnitPrice	OrderStatus	LastModified
1011	2026-08-29	S001	P011	1	18999.00	Completed	2026-08-29
1051	2026-08-27	S001	P011	5	18999.00	Cancelled	2026-08-27
1081	2026-08-27	S001	P011	5	18999.00	Cancelled	2026-08-27
1021	2026-08-27	S001	P006	1	8999.00	Completed	2026-08-27

Step 11: Aggregate Daily Sales

- Added an Aggregate transformation (aggregate1), grouping by ProductID, StoreID, OrderDate, ProductName, StoreName and Category.

Microsoft Azure | Data Factory | datafactoryde265487

Factory Resources

- Pipelines
 - PL_03_Load_CuratedZone
 - PL_01_Ingest_RawZone
 - PL_02_Transform_ProcessedZone
- Datasets
 - AzureSqlTable1
 - AzureSqlTable1_Orders
 - DelimitedText1
 - DelimitedText2
 - DelimitedText3
 - DelimitedText_Stores
 - Orders
 - Products
 - RestResource_Products
 - Stores
- Data flows
 - dataflow1
- Power Query

Aggregate settings

Columns	Name as
orders@ProductID	ProductID
orders@StoreID	StoreID
OrderDate	OrderDate
ProductName	ProductName
StoreName	StoreName
Category	Category

- Defined the aggregate expressions: Ordercount = count(OrderID), TotalSales = sum(orders@UnitPrice) and TotalStock = sum(Quantity).

Microsoft Azure | Data Factory | datafactoryde265487

Factory Resources

- Pipelines
 - PL_01_Ingest_RawZone
- Datasets
 - AzureSqlTable1_Orders
 - DelimitedText1
 - DelimitedText2
 - DelimitedText3
 - DelimitedText_Stores
 - Orders
 - Products
 - RestResource_Products
 - Stores
- Data flows
 - dataflow1
- Power Query

Aggregate settings

Grouped by: ProductID, StoreID, OrderDate

Column	Expression
Ordercount	count(OrderID)
TotalSales	sum(orders@UnitPrice)
TotalStock	sum(Quantity)

- Previewed the aggregated output — 62 rows summarizing daily sales per store and product.

Factory Resources

- Pipelines
 - PL_03_Load_CuratedZone
 - PL_01_Ingest_RawZone
 - PL_02_Transform_ProcessedZone
- Datasets
 - AzureSqlTable1
 - AzureSqlTable1_Orders
 - DelimitedText1
 - DelimitedText2
 - DelimitedText3
 - DelimitedText_Stores
 - Orders
 - Products
 - RestResource_Products
 - Stores
- Data flows
 - dataflow1
- Power Query
 -

Data preview

ProductID	StoreID	OrderDate	ProductName	StoreName	Category	Ordercount
P001	S001	2026-08-27	Wireless Mouse	Kochi Central	Electronics	4
P001	S006	2026-08-27	Wireless Mouse	Pune Market	Electronics	2
P001	S001	2026-08-29	Wireless Mouse	Kochi Central	Electronics	1
P001	S006	2026-08-30	Wireless Mouse	Pune Market	Electronics	1

Step 12: Write to the Silver Zone

- Configured the sink (sink1), exporting the aggregated stream to the DelimitedText3 dataset in the processed container.

Factory Resources

- Pipelines
 - PL_01_Ingest_RawZone
 - PL_02_Transform_ProcessedZone
- Datasets
 - AzureSqlTable1_Orders
 - DelimitedText1
 - DelimitedText2
 - DelimitedText3
 - DelimitedText_Stores
 - Orders
 - Products
 - RestResource_Products
 - Stores
- Data flows
 - dataflow1
- Power Query
 -

Sink configuration

- Output stream name: sink1
- Description: Export data to DelimitedText3
- Incoming stream: aggregate1
- Sink type: Dataset
- Dataset: DelimitedText3

Step 13: Run and Verify the Transform Pipeline

- Ran Debug on PL_02_Transform_ProcessedZone — the Data flow activity succeeded.

The screenshot displays the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane shows a list of pipelines, including 'PL_02_Transform_ProcessedZone'. The main area shows the 'dataflow1' pipeline run details. A red box highlights the 'Data flow' activity in the pipeline canvas and the 'Output' tab below it. The 'Output' tab shows a table with columns: Activity name, Activity status, Activity type, Run start, Duration, Integration runtime, and User properties. The row for 'dataflow1' shows a 'Succeeded' status.

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties
dataflow1	Succeeded	Data flow	9/8/2026, 12:06:33 PM	2m 1s	debugpool-8cores-General-70 (East US)	

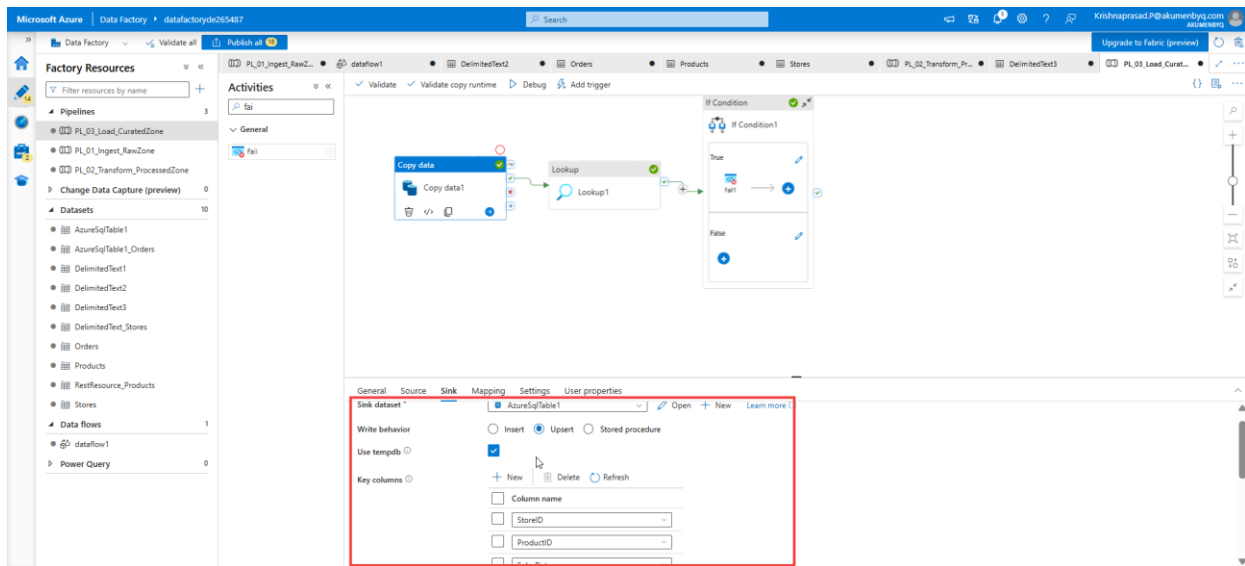
- Verified the processed container: the curated, joined and aggregated CSV (ProductID, StoreID, OrderDate, ProductName, StoreName, Category, Ordercount, TotalSales, TotalQuantity) had been written to the Silver zone.

The screenshot shows the Microsoft Azure Blob Storage interface. The left pane displays the 'processed' container. The main area shows a preview of a CSV file named 'part-00000-c65eb804-dea4-40f5-a83d-685d9214c84a-c000.csv'. A red box highlights the CSV content, which includes columns: ProductID, StoreID, OrderDate, ProductName, StoreName, Category, Ordercount, TotalSales, TotalQuantity. The data rows show various product details and their sales/quantity information.

ProductID	StoreID	OrderDate	ProductName	StoreName	Category	Ordercount	TotalSales	TotalQuantity
P010	S010	2026-08-28	Wireless Earbuds	Ahmedabad Hub	Audio	1	2999	5
P012	S002	2026-08-30	Gaming Mouse	Bangalore Mall	Electronics	1	3999	4
P011	S006	2026-08-27	27-inch Monitor	Pune Market	Electronics	2	37998	2
P002	S002	2026-08-30	Mechanical Keyboard	Bangalore Mall	Electronics	1	2499	2
P005	S005	2026-08-29	Bluetooth Speaker	Mumbai High Street	Audio	1	3499	5
P013	S003	2026-08-29	HDMI Cable	Chennai Central	Accessories	2	1798	2
P002	S007	2026-08-27	Mechanical Keyboard	Delhi City Mall	Electronics	1	2499	2
P005	S005	2026-08-31	Bluetooth Speaker	Mumbai High Street	Audio	3	18497	15
P009	S009	2026-08-29	Smart Watch	Kolkata Point	Wearables	1	4999	4
P003	S008	2026-08-28	USB-C Cable	Jaipur Square	Accessories	1	699	3
P007	S007	2026-08-27	Webcam HD	Delhi City Mall	Electronics	1	2299	2
P005	S010	2026-08-31	Bluetooth Speaker	Ahmedabad Hub	Audio	2	6998	4
P006	S006	2026-08-28	Noise Cancelling Headphones	Pune Market	Audio	3	26997	18
P004	S004	2026-08-28	Laptop Stand	Hyderabad Plaza	Accessories	1	1299	4
P014	S009	2026-08-30	Tablet	Kolkata Point	Electronics	2	31998	10
P011	S001	2026-08-29	27-inch Monitor	Kochi Central	Electronics	1	18999	1
P001	S001	2026-08-29	Wireless Mouse	Kochi Central	Electronics	1	499	1
P011	S006	2026-08-28	27-inch Monitor	Pune Market	Electronics	1	18999	1
P003	S003	2026-08-27	USB-C Cable	Chennai Central	Accessories	1	699	3

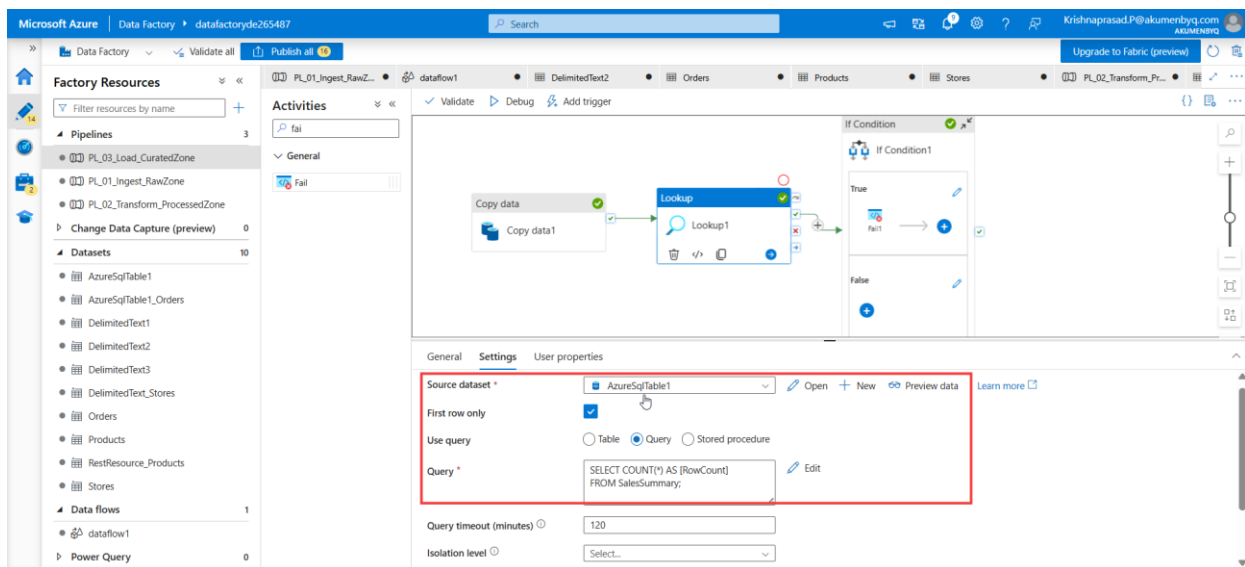
Step 14: Build PL_03_Load_CuratedZone — Upsert to Gold

- Created pipeline PL_03_Load_CuratedZone with a Copy data activity (Copy data1) sinking into AzureSqlTable1 with Write behavior set to Upsert (via a temp table) and key columns including StoreID and ProductID, so reruns update existing rows instead of duplicating them.

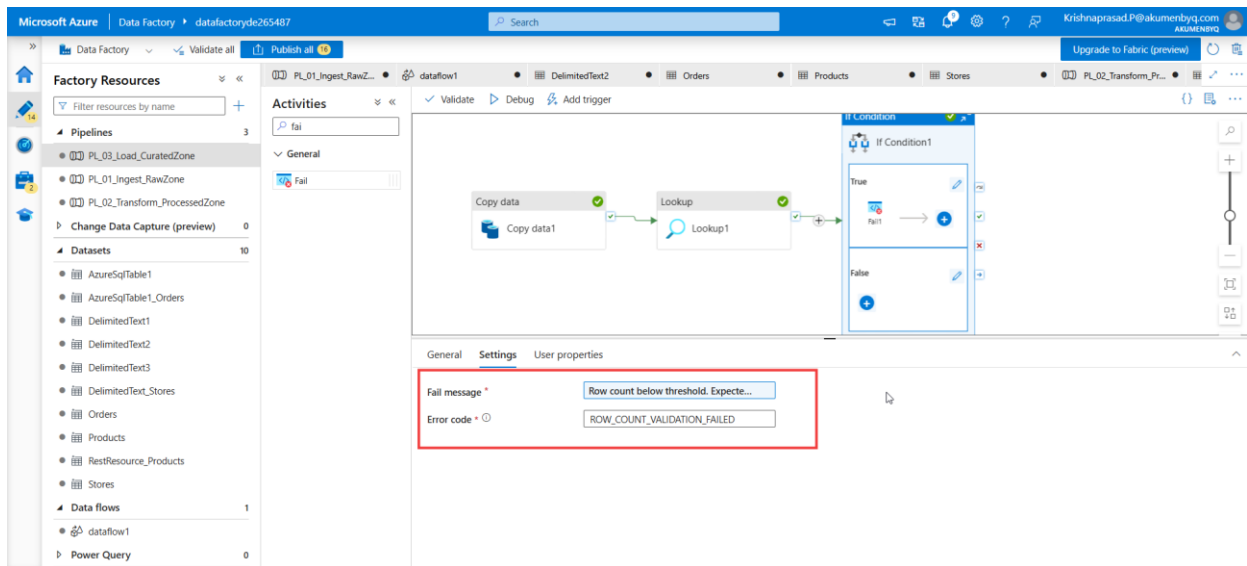


Step 15: Validate the Row Count

- Added a Lookup activity (Lookup1) after the copy, running `SELECT COUNT(*) AS [RowCount] FROM SalesSummary` to check how many rows landed in the curated table.

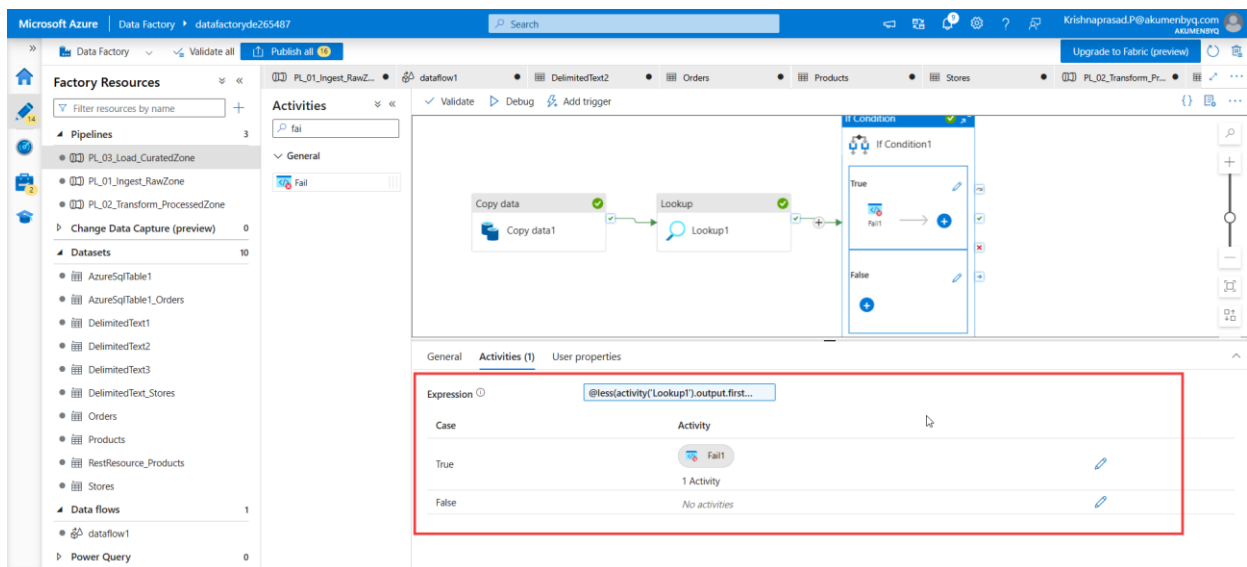


- Added a Fail activity (Fail1) with the message "Row count below threshold..." and error code `ROW_COUNT_VALIDATION_FAILED`, ready to stop the pipeline if the load didn't meet expectations.



Step 16: Branch on the Validation Result

- Added an If Condition activity evaluating `@less(activity('Lookup1').output.first...)` — routing to Fail1 on True, and taking no action on False when the row count is healthy.



- Ran Debug on PL_03_Load_CuratedZone — If Condition1, Lookup1 and Copy data1 all succeeded, confirming the row count passed validation.

Pipeline run ID: b598ce31-68ff-4958-b815-2ed9cf40811c **Pipeline status:** Succeeded

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...
If Condition1	Succeeded	If Condition	9/8/2026, 1:00:16 PM	Less than 1s		
Lookup1	Succeeded	Lookup	9/8/2026, 12:59:48 PM	27s	AutoResolveIntegrationRuntime (East US)	
Copy data1	Succeeded	Copy data	9/8/2026, 12:58:59 PM	48s	AutoResolveIntegrationRuntime (East US)	

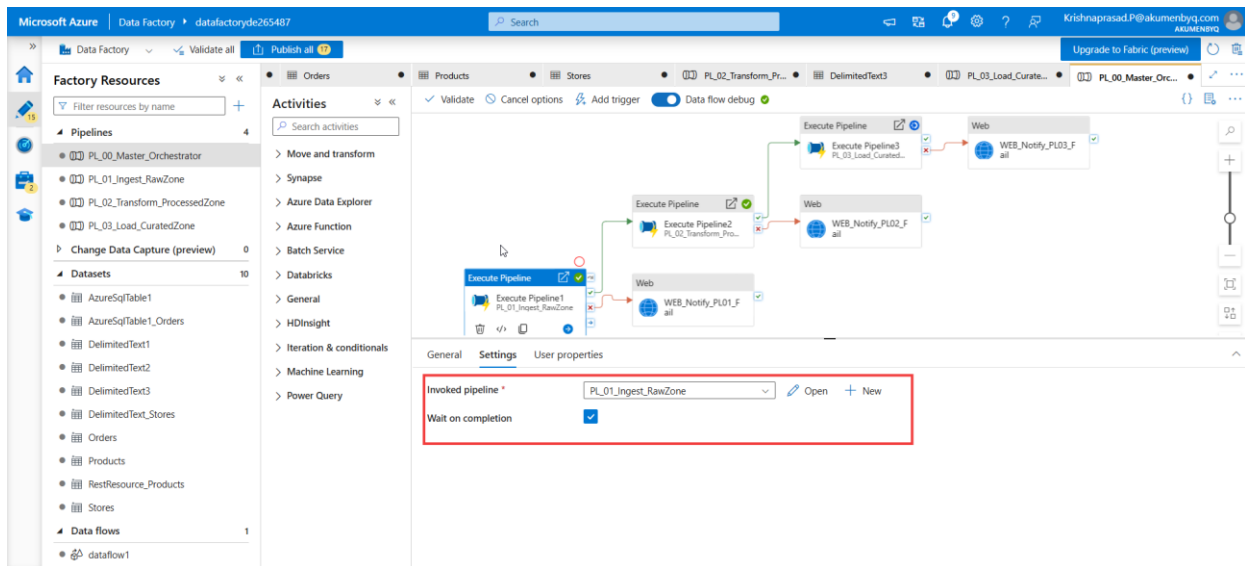
- Queried the curated SalesSummary table directly — 62 rows of business-ready sales data (SalesDate, StoreID, StoreName, ProductID, ProductName, Category, TotalQuantity, TotalSales) ready for Power BI.

Query: SELECT TOP (1000) * FROM [dbo].[SalesSummary]

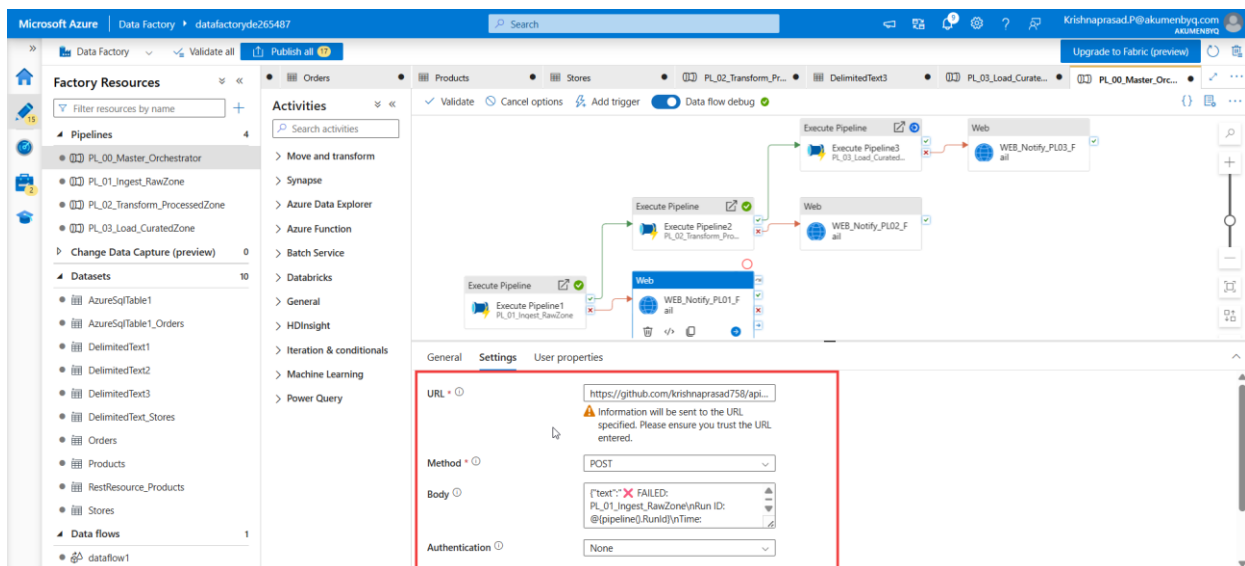
	SalesDate	StoreID	StoreName	ProductID	ProductNa...	Category	TotalQuantity	TotalSales
1	2026-08-27	S001	Kochi Central	P001	Wireless Mouse	Electronics	4	1996.00
2	2026-08-27	S001	Kochi Central	P006	Noise Cancellin...	Audio	7	26997.00
3	2026-08-27	S001	Kochi Central	P011	27-inch Monitor	Electronics	10	37998.00
4	2026-08-27	S003	Chennai Central	P003	USB-C Cable	Accessories	3	699.00
5	2026-08-27	S003	Chennai Central	P013	HDMI Cable	Accessories	3	899.00
6	2026-08-27	S005	Mumbai High St	P005	Bluetooth Spea...	Audio	5	3499.00

Step 17: Build PL_00_Master_Orchestrator

- Created the master pipeline PL_00_Master_Orchestrator with an Execute Pipeline activity (Execute Pipeline1) invoking PL_01_Ingest_RawZone with Wait on completion enabled.

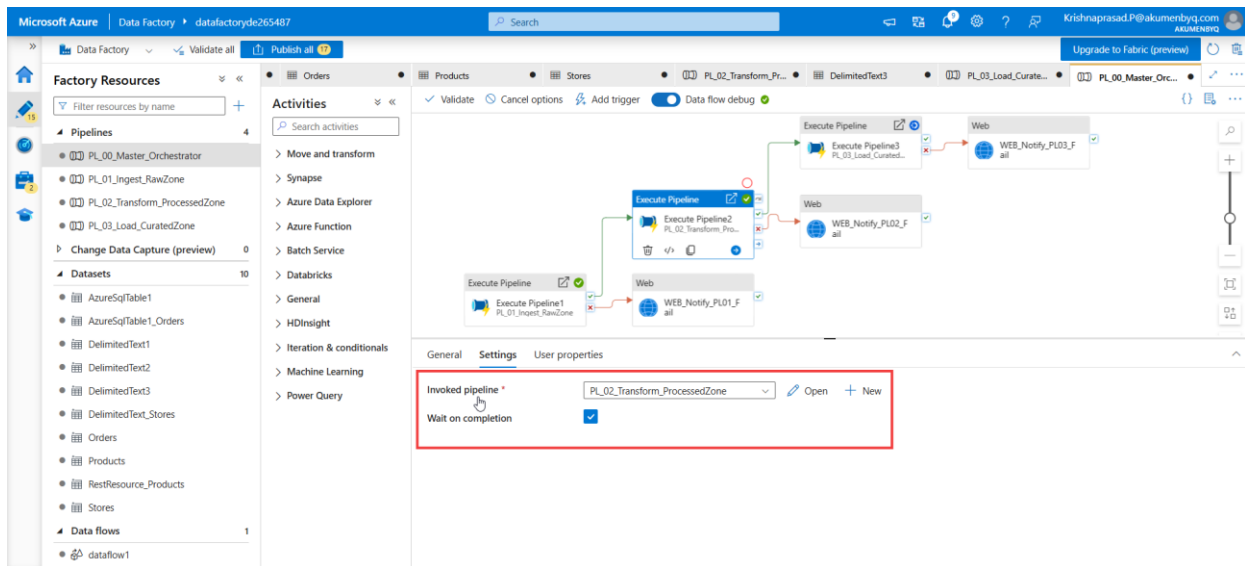


- Attached a Web activity (WEB_Notify_PL01_F) on the failure path, posting a JSON payload with the failed pipeline name, run ID and time to a GitHub webhook endpoint.

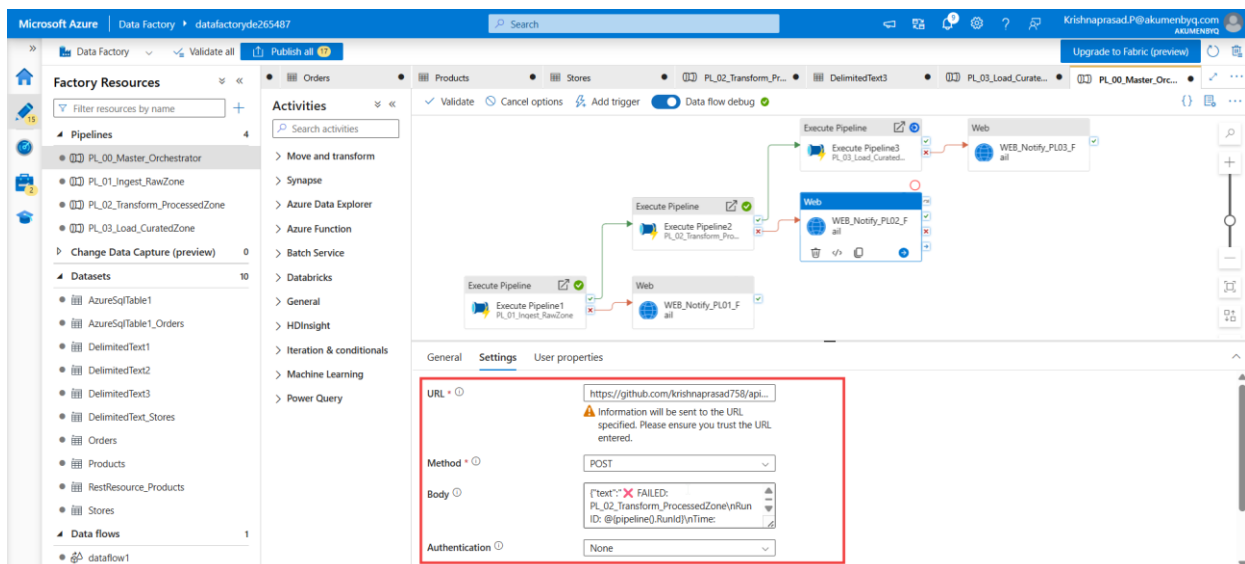


Step 18: Chain the Transform Pipeline

- Added Execute Pipeline2, invoking PL_02_Transform_ProcessedZone on success of the first step, with Wait on completion enabled.

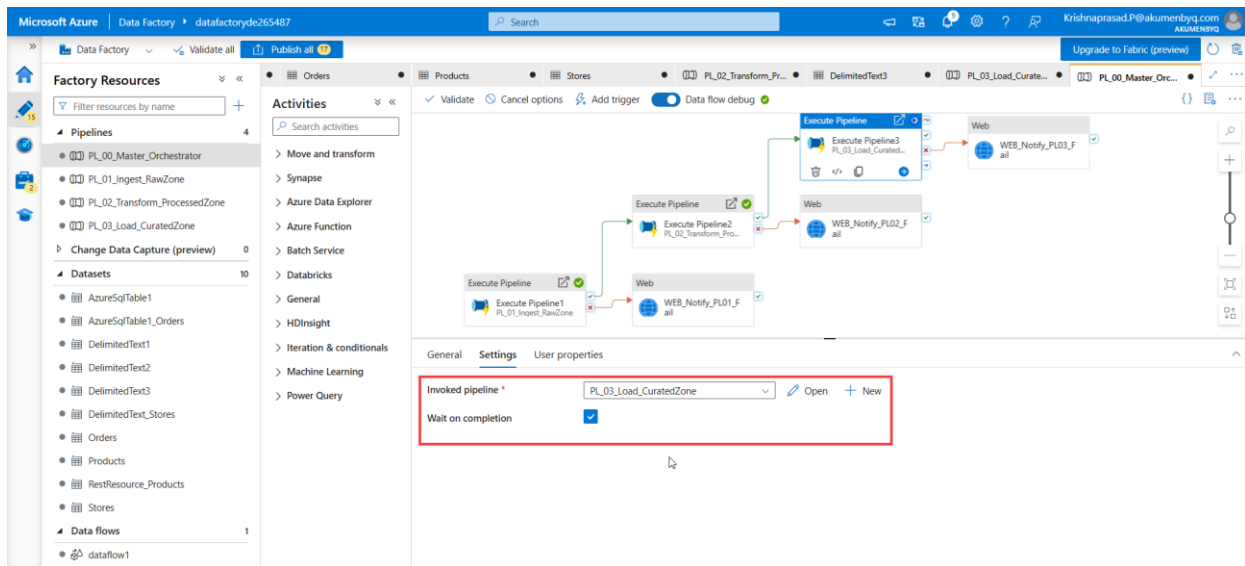


- Attached WEB_Notify_PL02_F on its failure path, posting the same style of failure notification for PL_02.

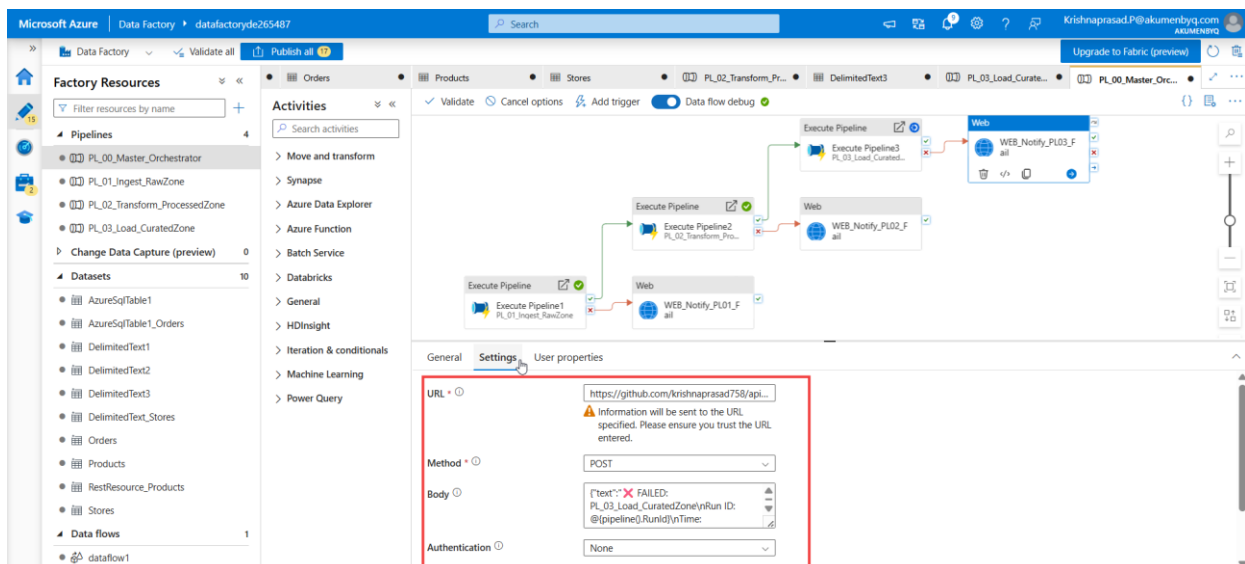


Step 19: Chain the Load Pipeline

- Added Execute Pipeline3, invoking PL_03_Load_CuratedZone on success of the transform step, with Wait on completion enabled.



- Attached WEB_Notify_PL03_F on its failure path, completing the notification coverage across all three stages.



Step 20: Run the End-to-End Orchestration

- Ran Debug on PL_00_Master_Orchestrator — Execute Pipeline1, Execute Pipeline2 and Execute Pipeline3 all succeeded in sequence, confirming the full Bronze → Silver → Gold flow runs end-to-end from a single triggered pipeline.

Microsoft Azure | Data Factory | datafactoryde265487

Search

Krishnaprasad.P@akumenbyg.com

Factory Resources

Filter resources by name

Pipelines4

PL_00_Master_Orchestrator

PL_01_Ingest_RawZone

PL_02_Transform_ProcessedZone

PL_03_Load_CuratedZone

Change Data Capture (preview)0

Datasets10

AzureSqlTable1

AzureSqlTable1_Orders

DelimitedText1

DelimitedText2

DelimitedText3

Orders

Products

RestResource_Products

Stores

Data flows1

dataflow1

Activities

Search activities

Move and transform

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Products

Stores

PL_02_Transform_Pr...

DelimitedText3

PL_03_Load_Curate...

PL_00_Master_Orc...

Upgrade to Fabric (preview)

Validate

Cancel options

Add trigger

Data flow debug

Execute Pipeline

Execute Pipeline1

Execute Pipeline2

Execute Pipeline3

Web

WEB_Notify_PL01_F

WEB_Notify_PL02_F

WEB_Notify_PL03_F

Parameters

Variables

Settings

Output

Pipeline run

ID

241f1eb6-6d2d-44a3-91d8-c51484b99386

Pipeline status

In progress

All status

Showing 1 - 3 of 3 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...
Execute Pipeline3	Succeeded	Execute Pipelin	9/8/2026, 1:26:57 PM	1m 16s		
Execute Pipeline2	Succeeded	Execute Pipelin	9/8/2026, 1:24:42 PM	2m 15s		
Execute Pipeline1	Succeeded	Execute Pipelin	9/8/2026, 1:23:35 PM	1m 6s		

Pipeline run ID 241f1eb6-6d2d-44a3-91d8-c51484b99386 Pipeline status In progress						
All status						
Showing 1 - 3 of 3 items						
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...
Execute Pipeline3	Succeeded	Execute Pipelin	9/8/2026, 1:26:57 PM	1m 16s		
Execute Pipeline2	Succeeded	Execute Pipelin	9/8/2026, 1:24:42 PM	2m 15s		
Execute Pipeline1	Succeeded	Execute Pipelin	9/8/2026, 1:23:35 PM	1m 6s		